

SKILLS & TOOLING MATRIX FOR COMMON SOFTWARE ROLES

A Professional Guide to Technologies, Prerequisites, History, and Career Paths

THE ARCHITECT'S PLAN



Section 1: The Master Plan

- Our Methodology & The Big Picture
- The Consolidated Tooling Matrix



Section 2: Role Blueprints

- Detailed schematics for eight core software development roles.



Section 3: Engineering Schematics & Codes

- Foundational Practices: DevOps, Testing & CI/CD
- Project Structure & Architecture Templates
- Professional Practice: Ethics, Security & Accessibility



Section 4: Commencement of Work

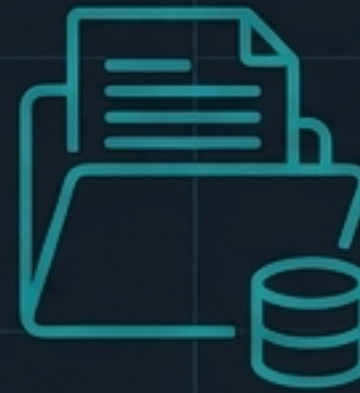
- Career Acceleration & Your Next Actions

METHODOLOGY: FROM IMAGE TO INSIGHT



Extract & Map

Core roles and technologies were identified and translated from the source visual. A small, semi-transparent version of the source image is shown behind this step's icon.



Categorize & Enrich

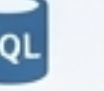
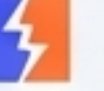
Technologies were grouped into a standardized taxonomy (Languages, Frameworks, Tools). Each role was enriched with prerequisites, history, project ideas, and career roadmaps.



Validate & Structure

The complete dataset was cross-referenced with industry standards and structured for professional use.

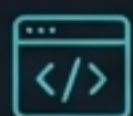
THE CONSOLIDATED TOOLING MATRIX

Role	Core Languages	Key Frameworks/Libraries	Essential Tools/Platforms
 Backend Dev	     Python, Node.js, Java, Go, C#	e.g., Express, Spring	  MongoDB, Docker
 Frontend Dev	  JavaScript, TypeScript	   React, Angular, Tailwind CSS	  HTML5, CSS3
 Mobile Dev	   Swift, Kotlin, Java	  Flutter, React Native	  Xcode, Android SDK
 Game Dev	  C#, C++	  Unity, Unreal Engine	  Blender, Git
 Security Specialist	   Python, Bash, SQL	N/A	    Linux, Nmap, Wireshark, Burp Suite
 Full-Stack Dev	   JavaScript, TypeScript, Node.js	 React	   PostgreSQL, Docker, Git
 Data Analyst	  SQL, Python	 Pandas	    Excel, Power BI, Tableau, Looker Studio
 AI/ML Engineer	 Python	    NumPy, Pandas, PyTorch, TensorFlow	  Jupyter, Docker

BLUEPRINT: BACKEND DEVELOPER

Designs and implements server-side logic, APIs, and data persistence

CORE COMPETENCIES



Languages: Python, Node.js, Java, Go, C# (#94A3B8)



Databases/Infra: MongoDB, PostgreSQL (#94A3B8)



Tools: Docker (#94A3B8)



Prerequisites:

Annonataiments (#94A3B8)

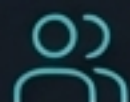
- ✓ Algorithms & Data Structures
- ✓ Networking Basics
- ✓ Linux/CLI Fundamentals
- ✓ Database Theory

CAREER PATH & HISTORY



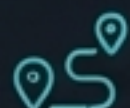
Historical Context:

- Origins in CGI scripts (mid-90s)
- Rise of server-side frameworks (Django/Rails, 2005-)
- Era of microservices & containers (2010s)

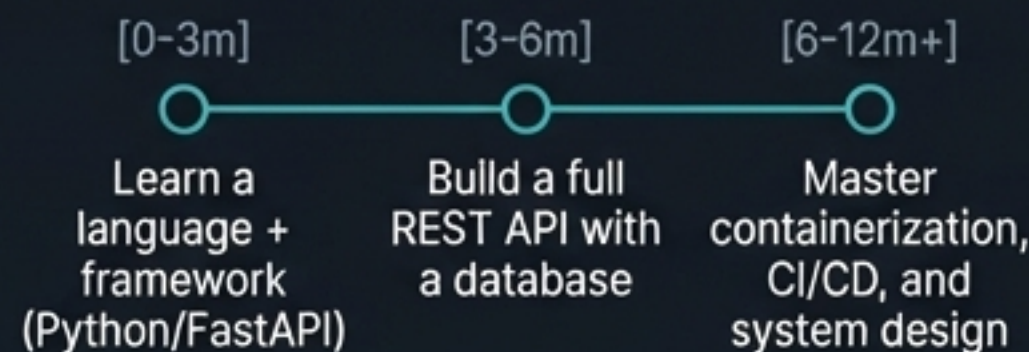


Key People:

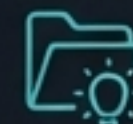
- Guido van Rossum (Python)
- Ryan Dahl (Node.js)



Roadmap:

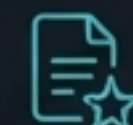


PORTFOLIO & IMPACT



Project Idea: Personal Budget API

- **Scope:** REST API for expense tracking with user auth.
- **Tech:** FastAPI, MongoDB, Docker, JWT.
- **Deliverables:** OpenAPI docs, Postman collection, unit tests.



Resume Bullet:

Designed and implemented a RESTful API using FastAPI and PostgreSQL; containerized with Docker and deployed via CI/CD, improving response time by 30% through query optimization.

BLUEPRINT: FRONTEND DEVELOPER

Builds interactive and accessible user interfaces for the web.

CORE COMPETENCIES

 **Markup/Styling**
HTML, CSS, Tailwind CSS

 **Languages/Frameworks**
JavaScript, TypeScript, React, Angular

 **Tools**
Git, Package Managers (npm/yarn)

 **Prerequisites**

- ✓ UX/UI Design Principles
- ✓ Browser APIs & DOM
- ✓ Accessibility (WCAG)
- ✓ Responsive Design

CAREER PATH & HISTORY

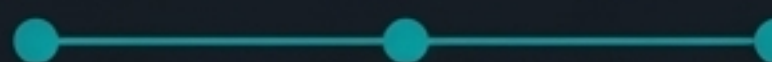
 **Historical Context**

- Evolution from static HTML pages to DHTML (late 90s)
- Rise of JS libraries (jQuery, 2006)
- SPA framework era (AngularJS/React, 2010s)

 **Key People**

- Brendan Eich (JavaScript)
- Jordan Walke (React/Meta)

 **Roadmap**



0-3m
Master HTML/CSS/JS fundamentals

3-6m
Build projects with a modern framework like React

6-12m+
Learn TypeScript, state management, and testing

PORTFOLIO & IMPACT

 **Project Idea: Design System Playground**

- Scope: Component library with responsive layouts and accessibility checks.
- Tech: React, TypeScript, Tailwind CSS, Storybook.
- Deliverables: Live demo site, documented components.

 **Resume Bullet** ☆

Built a responsive and accessible component library using React, TypeScript, and Tailwind CSS; documented all components in Storybook, improving developer efficiency by 40%.

BLUEPRINT: MOBILE DEVELOPER

Develops native or cross-platform applications for mobile devices.



CORE COMPETENCIES



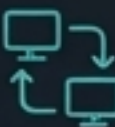
Native iOS

Swift, SwiftUI, Xcode



Native Android

Kotlin, Java, Android SDK



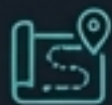
Cross-Platform

Flutter, React Native



Prerequisites

- ✓ Mobile UI/UX Patterns
- ✓ Understanding of Memory Management
- ✓ API Integration
- ✓ Platform-specific Guidelines (HIG/Material Design)



CAREER PATH & HISTORY



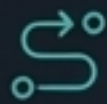
Historical Context

- Pre-smartphone era (J2ME/Symbian)
- Apple's App Store launch (2008)
- Android's rise (2009-), emergence of cross-platform solutions (2015-)



Key Organizations

- Apple
- Google



Roadmap



0-3m

Choose a platform (iOS/Android/Cross) and learn the language



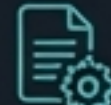
3-6m

Build a CRUD app with local storage



6-12m+

Integrate CI/CD, advanced features, and publish to an app store



PORTFOLIO & IMPACT

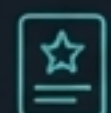


Project Idea: Habit Tracker App

Scope: CRUD tasks, reminders, local storage sync, simple analytics.

Tech: Flutter or Swift/Kotlin, local SQLite database.

Deliverables: Build instructions, UX screenshots, source code.



Resume Bullet

"Developed a cross-platform habit-tracking app in Flutter with local persistence and push notifications; integrated automated tests and a CI build pipeline."

BLUEPRINT: GAME DEVELOPER

Produces interactive games and simulations using game engines and 3D tools.

CORE COMPETENCIES

Languages

C#, C++

Engines

Unity, Unreal Engine

3D/Assets

Blender, Maya, 3ds Max

Tools

Git (specifically Git LFS for large assets)

Prerequisites

- ✓ Strong Math (Linear Algebra, Trigonometry)
- ✓ Physics Fundamentals
- ✓ Game Design Principles
- ✓ 3D Pipeline Knowledge

CAREER PATH & HISTORY

Historical Context

- Arcade era (70s-80s), rise of 3D graphics (90s)
- Democratization with engines like Unity (2005-)
- Indie game boom (2010s)

Key People

Tim Sweeney (Epic Games/Unreal Engine)
John Carmack (id Software/Doom)

Roadmap

0-3m



Learn an engine (Unity/C#) and build simple prototypes

3-6m



Complete a small game loop with art assets

6-12m+



Tackle complex systems like AI, networking, and optimization

PORTFOLIO & IMPACT

Project Idea: Small 2D Platformer

Scope: One playable level with character controls, basic enemy AI, and a scoring system.

Tech: Unity (C#), Git LFS.

Deliverables: Playable build, design document, notes on asset pipeline.

Resume Bullet ★

Created a 2D platformer in Unity with a custom player controller and enemy AI; managed all assets and version control using Git LFS.

BLUEPRINT: SECURITY SPECIALIST / ETHICAL HACKER

Assesses and strengthens system and network security through vulnerability analysis and penetration testing.

CORE COMPETENCIES

OS/Scripting: Linux, Bash, Python

Network Tools: Nmap, Wireshark

Web Tools: Burp Suite

Databases: SQL (for understanding injections)

Prerequisites

- ✓ Deep Networking Knowledge (TCP/IP)
- ✓ Operating System Internals
- ✓ Web Application Architecture
- ✓ Cryptography Basics

CAREER PATH & HISTORY

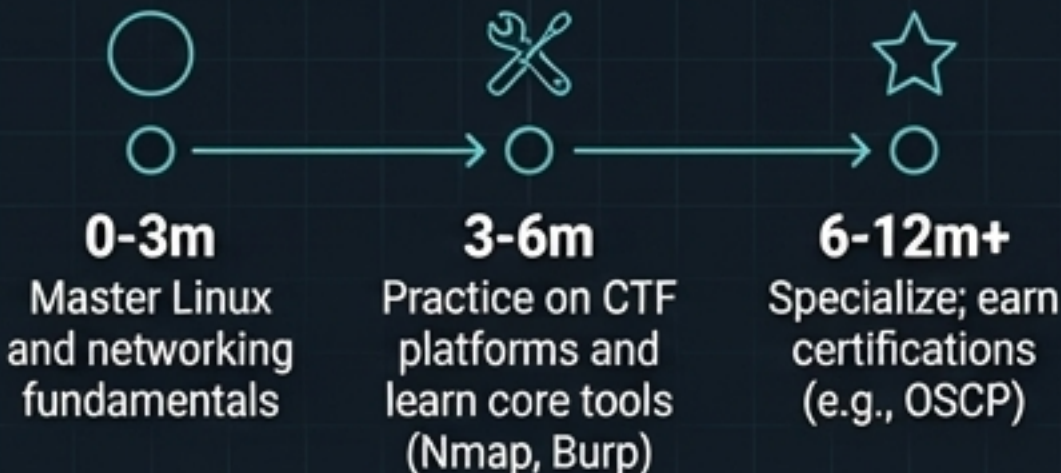
Historical Context

- Origins in "phreaking" and early hacking culture (70s-80s)
- Formalization of "ethical hacking" (90s)
- Rise of bug bounty programs (2010s)

Ethical Mandate

All work must be guided by strict ethical codes and responsible disclosure policies.

Roadmap



PORTFOLIO & IMPACT

Project Idea: Web App Pentest Report

Scope: Assess a deliberately vulnerable demo application (e.g., DVWA).

Tech: Kali Linux, Nmap, Burp Suite.

Deliverables: A professional vulnerability report with findings, impact analysis, and remediation suggestions.

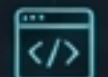
Resume Bullet

Performed network and web application assessments using Nmap, Burp Suite, and Wireshark; produced mitigation-focused reports and prioritized remediation steps for stakeholders.

BLUEPRINT: FULL-STACK DEVELOPER

Combines frontend and backend skills to build and deploy end-to-end web applications.

CORE COMPETENCIES

 **Frontend**
JavaScript, TypeScript, React

 **Backend**
Node.js

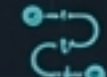
 **Database**
PostgreSQL

 **DevOps**
Docker, Git

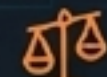
 **Prerequisites**

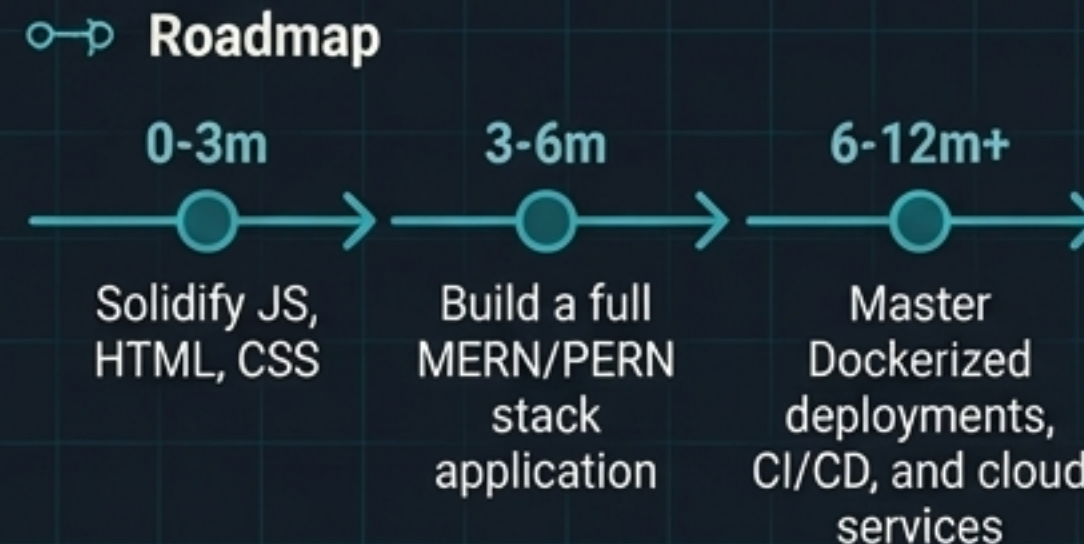
- ✓ Solid understanding of client and server architecture
- ✓ REST API design
- ✓ Database schema design
- ✓ Deployment processes

CAREER PATH & HISTORY

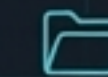
 **Historical Context**

- Gained popularity with MEAN/MERN stacks (2010s)
- Driven by single-language (JavaScript) ecosystems maturing

 **Key Trade-off**
Balances depth in one area with breadth across the entire application lifecycle.



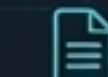
PORTFOLIO & IMPACT

 **Project Idea: Job Board Application**

Scope: A multi-tenant job board with user roles, search functionality, and an admin panel.

Tech: React, Node.js, PostgreSQL, Docker Compose.

Deliverables: Deployment instructions, architecture notes.

 **Resume Bullet**

Delivered an end-to-end job board web application with a React frontend, Node.js backend, and PostgreSQL database, all deployed via a containerized Docker workflow.

BLUEPRINT: DATA ANALYST

Extracts, cleans, interprets, and visualizes data to provide business insights



CORE COMPETENCIES

Languages/Libraries: SQL, Python, Pandas

Spreadsheets: Excel

BI Tools: Power BI, Tableau, Looker Studio

Prerequisites

- ✓ Statistics Fundamentals
- ✓ Data Cleaning Techniques
- ✓ Business Acumen/Domain Knowledge
- ✓ Data Storytelling



CAREER PATH & HISTORY

Historical Context

- Evolved from business intelligence roles
- Rise of user-friendly BI tools (Tableau, 2003)
- Python (Pandas, 2008) for data manipulation



Key Skill: The ability to translate technical findings into a compelling narrative for non-technical stakeholders."



Roadmap

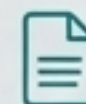


PORTFOLIO & IMPACT



Project Idea: Interactive Sales Dashboard

- **Scope:** Ingest sales CSV data, perform ETL, and create dashboards with key KPIs.
- **Tech:** Python (Pandas), SQL, Power BI/Tableau.
- **Deliverables:** Interactive dashboard, data cleaning notebook, executive summary write-up.



Resume Bullet

"Produced interactive sales dashboards using Python (Pandas) and Power BI, delivering weekly insights that directly informed revenue strategy and marketing decisions."

BLUEPRINT: AI / MACHINE LEARNING ENGINEER

Builds, trains, evaluates, and deploys machine learning models into production systems.

CORE COMPETENCIES

Languages/Libraries

Python, NumPy, Pandas

Frameworks

PyTorch, TensorFlow

Tools

Jupyter, Docker

Prerequisites

- ✓ Strong Math (Linear Algebra, Calculus, Probability)
- ✓ Statistics & Modeling
- ✓ Software Engineering Principles
- ✓ Data Structures & Algorithms

CAREER PATH & HISTORY

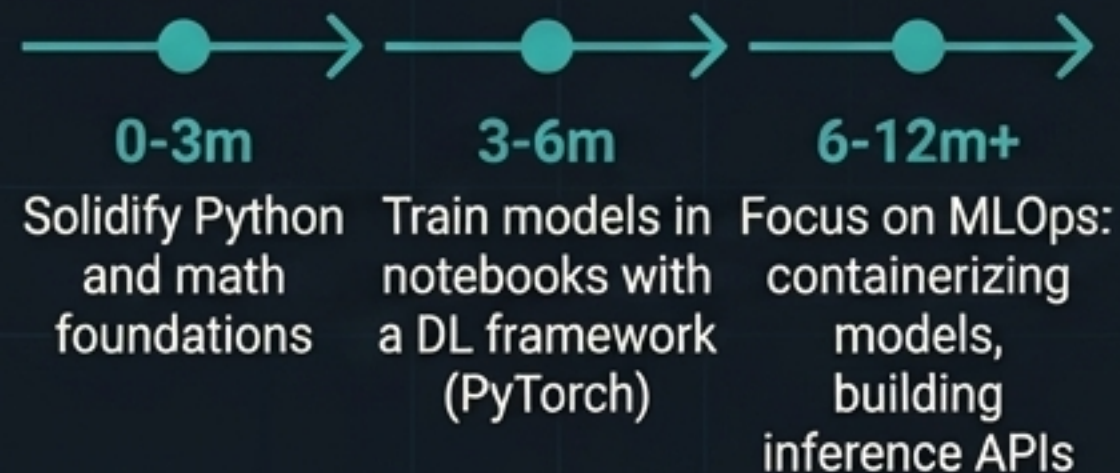
Historical Context

- Exploded after "ImageNet moment" (2012)
- Enabled by frameworks like TensorFlow (2015) and PyTorch (2016)

Key Organizations

- Google (TensorFlow)
- Meta (PyTorch)

Roadmap



PORTFOLIO & IMPACT

Project Idea

Image Classification Notebook & API

- **Scope:** Dataset prep, model training (transfer learning), evaluation, and a simple deployment API.
- **Tech:** Python, PyTorch/TF, Jupyter, Docker, FastAPI.
- **Deliverables:** Explained notebook, model weights, evaluation metrics, Dockerfile for API.

Resume Bullet

- Trained and evaluated CNN models for image classification using PyTorch; demonstrated deployment readiness via a lightweight FastAPI inference API in a Docker container.

THE ENGINEERING FOUNDATION: DEVOPS, TESTING & CI/CD



DevOps & Containerization

Concept: The practice of unifying software development (Dev) and IT operations (Ops) to shorten the development lifecycle.

Key Tool: Docker: Package applications and dependencies into isolated containers for consistency across environments. A non-negotiable modern skill.



Quality Assurance & Testing

Concept: Writing code to verify that your application code works as expected. Essential for maintainability and reliability.

Types: Unit Tests (test individual functions), Integration Tests (test how components work together), End-to-End Tests (simulate user flows).



Continuous Integration & Delivery (CI/CD)

Concept: Automating the build, test, and deployment process.

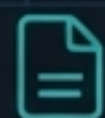
Key Tool: GitHub Actions / GitLab CI: Automatically run tests and deploy code upon commits, ensuring code quality and rapid iteration.

THE PROJECT BLUEPRINT: REPOSITORY & ARCHITECTURE

Canonical Repository Structure

A well-organized project is a well-understood project.

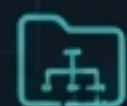
```
project-name/  
├── README.md  
├── Dockerfile  
├── .github/workflows/  
├── src/  
├── notebooks/  
├── data/  
└── tests/
```



A clear **README.md** is your project's front door.



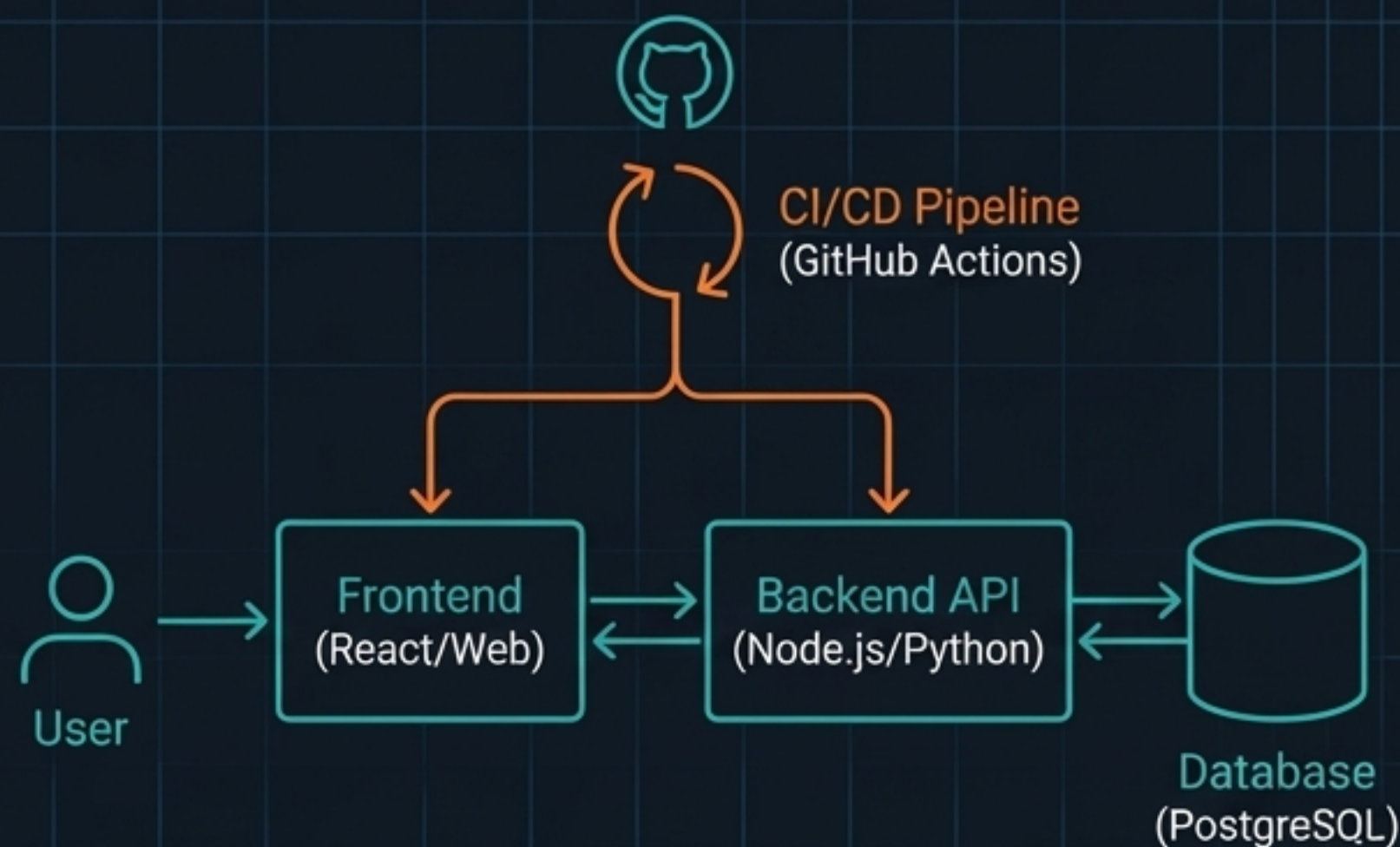
A **Dockerfile** ensures reproducibility.



Separate **src**, **tests**, and **notebooks** for clarity.

Generic Web App Architecture

Visualize your system before you build.



PROFESSIONAL PRACTICE: THE GUIDING CODES



Responsible Disclosure

Never publicly disclose a vulnerability without notifying the owner and allowing time for a fix. Always operate within a defined and authorized scope.



Handle Data with Care

For AI/ML, be aware of model bias and its societal impact. For all roles, respect user privacy (e.g., GDPR, CCPA) and minimize data collection.



Build for Everyone

Ensure applications are usable by people with disabilities. Use semantic HTML, provide alt text for images, and ensure sufficient color contrast (>4.5:1).



Know Your License

Use OSI-approved licenses. MIT for maximum permissiveness. Apache 2.0 if you want to require attribution. Understand the licenses of your dependencies.

CAREER ACCELERATION KIT: READINESS & RESOURCES

What Hiring Managers Look For

- ☐ **Clear README .md:** Explains the 'what, why, and how.'
- ☐ **Reproducible Setup:** Dockerfile or clear setup script.
- ☐ **Automated Tests:** Unit or integration tests are present.
- ☐ **CI Workflow:** A .github/workflows file shows professionalism.
- ☐ **Clean Code & Commits:** Demonstrates craftsmanship.
- ☐ **Live Demo (for web):** A link to a deployed version.

Where to Focus Your Learning

1. **Official Documentation:** The ultimate source of truth (e.g., React Docs, Python.org), Python.org).
2. **Project-Based Courses:** Platforms like Coursera, edX, or Udemy. Prioritize courses where you build something tangible.
3. **Foundational Books:** For timeless concepts like algorithms, system design, and security principles.
4. **Community & Open Source:** GitHub for code examples, **Stack Overflow** for specific problems.

COMMENCEMENT OF WORK: YOUR NEXT THREE ACTIONS



1 Choose Your Primary Role

Commit to one of the eight blueprints as your focus for the next 6-12 months. You can't master everything at once. Generalize later.



2 Build Three Portfolio-Grade Projects

One simple project to learn the basics. One medium project that mirrors a real-world application (like the ideas presented). One complex project that pushes your boundaries.



3 Document & Publish Your Work

A project that isn't documented and shared on GitHub or a blog might as well not exist. Follow the repository and README best practices.